

Aggregate functions and set operators

- The learning objectives for this week are:
 - Knowing what kind of query problems **aggregate functions** solve
 - Knowing how to use the aggregate functions `COUNT` , `SUM` , `AVG` , `MIN` and `MAX`
 - Knowing what kind of query problems **set operators** solve
 - Knowing how to combine result tables with `UNION` , `EXCEPT` and `INTERSECT` set operators

Aggregate functions

- Basic `SELECT` statement lets us retrieve specific columns from a table and filter rows with a `WHERE` clause
- However, when working with large datasets, we often need **summary information**, such as counts, averages, or totals, rather than individual rows
- That is, performing some calculation (such as sum or average) for **multiple rows** so that the end result is a **single summarized value** is a common query problem
- For example, *"what is the total number of courses?"*, or *"what is the sum of male teachers' salaries?"*
- Functions that perform such SQL operations are referred to as **aggregate functions**

COUNT aggregate function — counting rows

```
COUNT ( * | { [ DISTINCT ] column_expression } )
```

- The `COUNT` aggregate function returns the **total number of rows** that match the specified criteria
- The `COUNT` aggregate function and other aggregate functions are used to compute a value for a column in the `SELECT` statement

```
-- what's the number of courses in the Course table?  
SELECT COUNT(*) AS number_of_courses FROM Course
```

- The `COUNT` aggregate function operates on all the target table rows, leaving the result table with a single row

number_of_courses
7

Using aggregate functions with a WHERE clause

- We can also filter the rows the aggregate function operates on using the `WHERE` clause

```
-- what's the number of courses with more than 3 credits?  
SELECT COUNT(*) AS number_of_courses FROM Course  
WHERE credits > 3
```

- Now the aggregate function operates on the target table rows, **which match the `WHERE` clause condition**, leaving the result table with a single row

number_of_courses
2

Excluding NULL values with COUNT aggregate function

- We can also provide a column name for the `COUNT` aggregate function in which case the function returns the number of the **non-null values** of the given column

```
-- what's the number of students with an email address?  
SELECT COUNT(email) AS number_of_students_with_email  
FROM Student
```

Excluding NULL values with COUNT aggregate function

student_number	email
o354	0354@takkula.fi
o410	0410@takkula.fi
o473	NULL

```
-- count all rows
SELECT COUNT(*) AS total_number_of_students
FROM Student
-- total_number_of_students: 3

-- count rows with non-null email column value
SELECT COUNT(email) AS number_of_students_with_email
FROM Student
-- number_of_students_with_email: 2
```

SUM aggregate function — sum of column values

```
SUM ( [ DISTINCT ] column_expression )
```

- The **SUM** aggregate function returns the **sum of all the non-null values** of a column

```
-- what's the sum of salaries of female teachers?  
SELECT SUM(salary) AS sum_of_salaries FROM Teacher  
WHERE gender = 'F'
```

- The **SUM** aggregate function operates on all the target table rows, leaving the result table with a single row

sum_of_salaries
102273.00

AVG aggregate function — average of column values

```
AVG ( [ DISTINCT ] column_expression )
```

- The `AVG` aggregate function returns the **average of non-null values** of a column

```
-- what's the average grade from course with code "a730"?  
SELECT AVG(grade) AS average_grade FROM CourseGrade  
WHERE course_code = 'a730'
```

- The `AVG` aggregate function operates on all the target table rows, leaving the result table with a single row

average_grade
3

Preserving decimal values in average calculations

- Calculating the average includes a division operation, which can produce decimal numbers
- To avoid losing the decimal part of the average, we need to cast integer column values to a `DECIMAL` type

```
-- multiplying an integer with 1.0 will end up with a DECIMAL type
SELECT AVG(grade * 1.0) AS average_grade FROM CourseGrade
WHERE course_code = 'a730'
```

average_grade
3.333333

Rounding the average

- We can limit the number of decimal places in the result by using casting the result to a `DECIMAL` type with specific precision (the total number of decimal digits stored) and scale (the number of decimal digits stored to the right of the decimal point)

```
-- use scale of 2 in the DECIMAL type to round to two decimals places
SELECT CAST(AVG(grade * 1.0) AS DECIMAL(5, 2)) AS average_grade
FROM CourseGrade
WHERE course_code = 'a730'
```

`DECIMAL(5, 2)` $\Rightarrow$ $\underbrace{999}_{5 - 2 = 3 \text{ digits left}} . \underbrace{99}_{2 \text{ digits right}}$

average_grade
3.33

MIN aggregate function — the smallest column value

```
MIN ( column_expression )
```

- The `MIN` function returns the **smallest value** of a column

```
-- what's the lowest grade from course with code "a730"?  
SELECT MIN(grade) AS lowest_grade FROM CourseGrade  
WHERE course_code = 'a730'
```

- The `MIN` aggregate function operates on all the target table rows, leaving the result table with a single row

lowest_grade
1

MAX aggregate function — the largest column value

```
MAX ( column_expression )
```

- The `MAX` function returns the **largest value** of a column

```
-- what's the highest salary of male teachers?  
SELECT MAX(salary) AS highest_salary FROM Teacher  
WHERE gender = 'M'
```

- The `MAX` aggregate function operates on all the target table rows, leaving the result table with a single row

highest_salary
15780.00

Multiple aggregate functions in a single query

- We can have multiple aggregate functions in the same query

```
-- what's the highest and lowest grade from course with code "a730"?  
SELECT MAX(grade) AS highest_grade, MIN(grade) AS lowest_grade FROM CourseGrade  
WHERE course_code = 'a730'
```

- The result table contains a single row with two columns

highest_grade	lowest_grade
5	1

Only operating on distinct values

- The `COUNT`, `SUM` and `AVG` aggregate functions support the `DISTINCT` keyword for only operating on **distinct values**

course_code	instance_number	grade
a730	1	5
a730	1	3
a290	2	3

```
-- how many different grades have been given?  
SELECT COUNT(DISTINCT grade) AS number_of_different_grades FROM CourseGrade  
-- number_of_different_grades: 2
```

Restrictions of aggregate function columns

- If we use an aggregate function, we **can't include non-aggregate function columns** in the `SELECT` statement*

```
-- ✓ only aggregate function columns, all good here
```

```
SELECT COUNT(*) AS number_of_courses FROM Course
```

```
-- ✗ combination of aggregate function and non-aggregate function columns, this won't work
```

```
SELECT course_name, COUNT(*) AS number_of_courses FROM Course
```

- * That is, unless the non-aggregate function columns are included in a `GROUP BY` clause, but we will cover that later

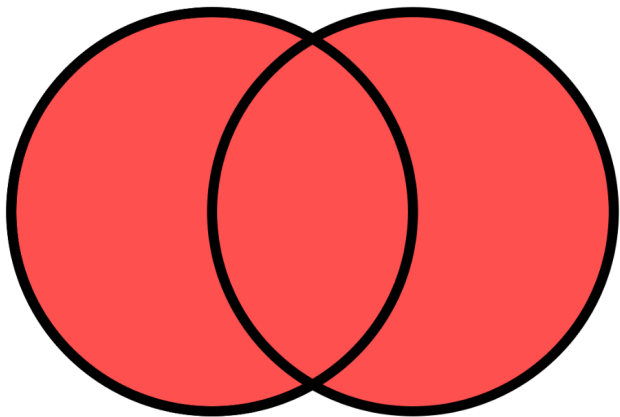
Restrictions of aggregate function columns

- If it would be possible, how would the RDMS know, which `course_name` value to display in the result table?

```
-- ❌ combination of aggregate function and non-aggregate function columns, this won't work  
SELECT course_name, COUNT(*) AS number_of_courses FROM Course
```

course_name	number_of_courses
???	2

Combining result tables with set operators



- We can use the results from **multiple result tables** using the `UNION`, `EXCEPT`, and `INTERSECT` **set operators**
- For example, the `UNION` operator returns **all** the rows from two or more result tables **without duplicate values**

```
-- what are all the surnames among teachers and students?  
SELECT surname FROM Teacher  
UNION  
SELECT surname FROM Student
```

UNION set operator — combine rows

"What are all the surnames among teachers and students?"

```
SELECT surname FROM Teacher
```

surname
Huhta
Hellerus

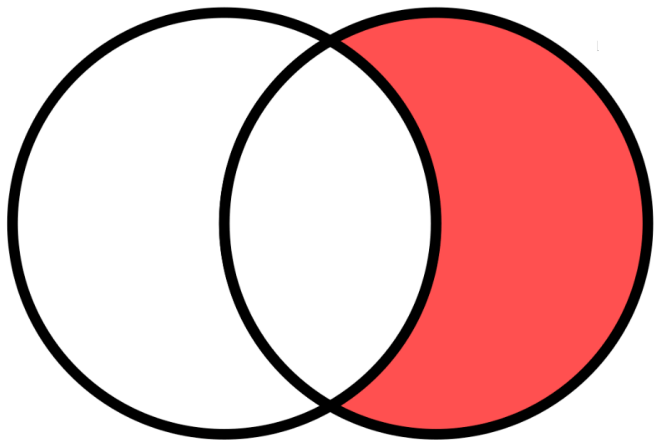
```
SELECT surname FROM Student
```

surname
Kokki
Kuikka

```
SELECT surname FROM Teacher  
UNION  
SELECT surname FROM Student
```

surname
Huhta
Hellerus
Kokki
Kuikka

EXCEPT set operator — exclude rows



- The `EXCEPT` operator returns only the rows from the first result table that are **not included** in the second result table **without duplicate values**

```
-- what are the campus cities that no student lives in?  
SELECT city FROM Campus  
EXCEPT  
SELECT city FROM Student
```

EXCEPT set operator — exclude rows

"What are the campus cities that no student lives in?"

```
SELECT city FROM Campus
```

city
Helsinki
Vantaa

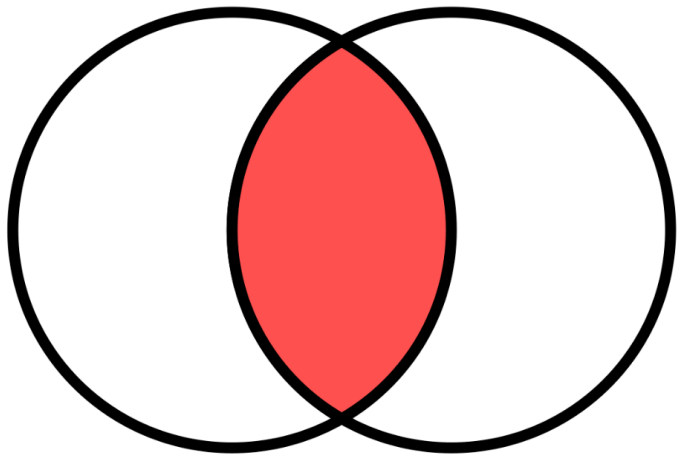
```
SELECT city FROM Student
```

city
Helsinki
Espoo

```
SELECT city FROM Campus  
EXCEPT  
SELECT city FROM Student
```

city
Vantaa

INTERSECT set operator — include common rows



- The `INTERSECT` operator returns only the rows that **exist in both** result tables **without duplicate values**

```
-- what are the campus cities  
-- that have students living in them?  
SELECT city FROM Campus  
INTERSECT  
SELECT city FROM Student
```

INTERSECT set operator — include common rows

"What are the campus cities that have students living in them?"

```
SELECT city FROM Campus
```

city
Helsinki
Vantaa

```
SELECT city FROM Student
```

city
Helsinki
Espoo

```
SELECT city FROM Campus
```

```
INTERSECT
```

```
SELECT city FROM Student
```

city
Helsinki

Restrictions of set operators

-  With set operators, the column names and data types of each `SELECT` statement **must match**

```
-- ✗ first_name column is missing from the latter SELECT statement.  
-- This will lead into an error.  
SELECT surname, first_name FROM Teacher  
UNION  
SELECT surname FROM Student
```

Summary

- We can perform calculations on multiple rows so that the end result is a single row using **aggregate functions**
- The `COUNT` aggregate function returns the **total number of rows**
- The `SUM` aggregate function returns the **sum of all the values** of a column
- The `AVG` aggregate function returns the **average of values** of a column
- The `MIN` aggregate function returns the **minimum value** of a column
- The `MAX` aggregate function returns the **maximum value** of a column
- The `UNION` , `EXCEPT` and `INTERSECT` **set operators** can be used to combine results of multiple result tables